Ian Douglas

Senior Platform Engineer | Cloud & Security Architect

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PROFESSIONAL SUMMARY

Platform and security engineer with 12+ years delivering secure, code-driven infrastructure for mission-critical services across Kubernetes, AWS, Azure, GCP, and bare metal. Ships platforms as versioned releases through Git-driven automation. Deep experience across the full lifecycle of identity and secrets infrastructure.

TECHNICAL PROFICIENCIES

Platforms & Containers: Kubernetes (EKS, AKS, GKE, self-managed control plane & etcd, bare metal), Docker, Custom Operators, Versioned Platform Releases, Linux Internals

Cloud: AWS (Organizations, multi-account, EC2, VPC, Transit Gateway, Route 53, IAM, KMS, Secrets Manager, Lambda), Azure (AKS, VMSS, Entra ID), GCP (GKE), Hybrid & Multi-Cloud Integration

Infrastructure as Code & Delivery: Terraform at scale (à la carte-as-code module design), GitOps (Argo CD, Flux CD), GitHub Actions, GitLab Pipelines, Jenkins, Git & Pull-Request Workflows

Software & Automation: Go (Golang), Python (Boto3, Pandas), Bash, PowerShell, SQL, API-Driven Automation, Agentic Development (Claude Code, Codex)

Identity & Security: HashiCorp Vault (PKI), SAML/Shibboleth SSO, Kerberos v5, OpenLDAP, Active Directory, Microsoft Entra ID, AWS IAM, AWS KMS, SOPS, Policy-as-Code (OPA), NIST SP 800-207 Zero Trust, Least-Privilege Design

Networking: Hybrid Connectivity, Transit Gateway, Site-to-Site VPN, VPC/VNet Peering, Hybrid DNS, IPv6, IPsec, Firewalls, Certificates & PKI

Observability & Data: Splunk, Prometheus, Grafana, Fluentd, Elasticsearch, Centralized Log Aggregation, PostgreSQL, MSSQL, High-Throughput Data Pipelines

PROFESSIONAL EXPERIENCE

The Motley Fool | Alexandria, VA (Remote)

Senior Platform Engineer (Contract) | 04/2026 – Present

- **CI/CD Migration Automation:** Built reusable Python tooling that migrated ~200 repositories from Jenkins to GitHub Actions, including moving secrets from AWS Secrets Manager to dotenvx, saving months of manual effort.
- **Cloud Infrastructure:** Operate production AWS and Kubernetes infrastructure.

- **AI-Accelerated Engineering:** Developed the migration toolchain with Claude Code agentic workflows, shipping production-grade automation in a fraction of the usual time.

Kings Mountain Security | Palo Alto, CA (Remote)

Co-Founder, DevSecOps Engineer & Security Architect | 01/2025 – Present

- **Zero Trust Architecture:** Co-founded a security consultancy delivering Terraform-based "Zero Trust Capability Packages" modeled on NIST SP 800-207, turning security requirements into deployable guardrails and technical controls.
- **Compliance Automation:** Developing Security-as-Code modules that automate DoD Zero Trust compliance across the infrastructure lifecycle.

FlowDelta Trading | Boulder, CO (Remote)

Senior Software Engineer, Python/Data (Contract) | 01/2025 – 04/2026

- **Backtesting Platform:** Developed a proprietary, multi-horizon analysis platform for a boutique trading desk, enabling rapid backtesting and validation of medium-frequency algorithms.
- **Data-Intensive Systems:** Engineered a vectorized Monte Carlo simulation engine to perform multi-dimensional parameter grid sweeps.

ZagTech | San Jose, CA (Remote)

Senior Platform Engineer | 04/2024 – 12/2024

- **À La Carte-as-Code:** Created a radically simple Terraform design for ordering complex Kubernetes clusters from a menu (Kate's Kitchen).
- **Fleet GitOps:** Automated Flux across multi-project, multi-environment cluster fleets with Terraform, Go, and Python, increasing deployment velocity while reducing change failure rates.
- **Commercial Platform:** Engineered a proprietary Kubernetes platform deployed across multiple enterprise clients; the codebase was a primary value driver in the firm's subsequent acquisition.
- **Cloud Networking:** Designed Azure network connectivity for client environments, including VNet peering, site-to-site VPN, and hub transit routing.
- **Cost Optimization:** Rightsized and tuned Azure infrastructure to meet client cost requirements.
- **Documentation & Runbooks:** Wrote implementation guides and operational runbooks for client handoff and internal teams.

Invitae Biosciences | San Francisco, CA (Remote)

Senior Platform Engineer | 07/2023 – 04/2024

- **Research Compute Platform:** Built a self-service platform on self-managed Kubernetes on AWS EC2 for petabyte-scale genomics workloads; executed zero-downtime cluster upgrades and stateful migrations.
- **Fleet GitOps:** Designed the Flux repository hierarchy and cluster bootstrap for the genomics platform.

- **Cluster Hardening:** Built an automated audit framework enforcing CIS Benchmarks and Pod Security Standards (Cluster Security Audit).
- **Secrets Management Architecture:** Designed least-privilege secrets management using SOPS, AWS KMS, and Shamir secret sharing.

Pearson | London, UK (Remote)

Senior Cloud Platforms Engineer | 09/2019 – 08/2021

- **Multi-Account AWS:** Built and operated platform infrastructure across a multi-account AWS environment managed with AWS Organizations.

- **GitOps Platform:** Built and operated self-service developer platforms on Kubernetes with GitOps-based continuous delivery, including ephemeral Kubernetes environments for development and testing.
- **Custom Kubernetes Operators:** Wrote Go-based operators to orchestrate AWS services from self-managed Kubernetes on EC2.
- **Centralized Logging:** Implemented log aggregation with a distributed pipeline (Fluentd, Python, AWS Lambda) feeding Elasticsearch and Splunk for real-time analysis.

Stanford University | Palo Alto, CA

DevOps Engineer & Security Architect | 12/2015 – 09/2019

- **Identity Modernization:** Partnered with Infrastructure teams to move the university's authentication and authorization stack (SAML/Shibboleth SSO, Kerberos v5, OpenLDAP) from legacy servers onto Kubernetes.

- **Zero-Trust Active Directory:** Architected AD whose domain controllers stay invisible until clients establish IPsec with ephemeral Vault certificates (paper).
- **Self-Service Cloud Identity:** Shipped it to production in Terraform over IPv6, so any VPC could join without IP coordination, peering, or NAT/Transit Gateway costs; scales to thousands of VPCs.
- **Hybrid Cloud Networking:** Built connectivity between on-premises infrastructure and AWS using Transit Gateway, site-to-site VPN, and VPC peering.
- **Multi-Account & Multi-Cloud:** Operated AWS Organizations multi-account environments and supported the university's growing GKE footprint alongside self-hosted Kubernetes.
- **Secrets Infrastructure:** Managed multi-tenant HashiCorp Vault clusters providing automated PKI, authentication proxies, and encryption as part of a three-person team.
- **Security Monitoring:** Integrated services with Splunk for centralized logging and security visibility.

Marvel Heroes (Gazillion Entertainment) | Foster City, CA

DevOps Engineer | 06/2014 – 12/2015

- **Cluster Ops Tooling:** Built PowerShell and SQL Server Management Objects (SMO) tooling to automate database operations and co-developed cluster operations tools for a flagship MMO with 40k+ daily active users, cutting service drain times by 93%.

- **Observability at Scale:** Operated large metrics and log aggregation servers for production game infrastructure.

Two Sigma | New York, NY

Systems Engineer | 05/2012 – 06/2014

- **High-Performance Fleet:** Automated bare-metal provisioning and configuration management for high-performance servers supporting research and trading operations.
- **Lifecycle Automation:** Orchestrated zero-touch deployment and lifecycle operations of 1,000+ secure workstations firm-wide.

Columbia University | New York, NY

Systems Engineer | 06/2010 – 05/2012

- **Infrastructure Operations:** Ran databases, ESX, storage, and VM infrastructure supporting Columbia's 22 libraries.

EDUCATION

University of Denver | Denver, CO

Bachelor of Arts (BA), Double Major: Mathematics & Philosophy

- **Mathematics:** Completed Bachelor of Science curriculum including grad-level coursework.
- **Philosophy:** Specialized in Formal Logic and Epistemology.